Statistical Modelling & Analysis (STAT2004/2904/7004)
Assignment 1

Submit your solutions via Blackboard as a single PDF file.

Due Friday 21 August at 4:00 pm

STAT2004/7004: Complete Exercises 1–4 for a maximum of 40 marks and a total of 10%.
STAT2904: Complete Exercises 1–5 for a maximum of 50 marks and a total of 10%.

1. (10 marks) The Rayleigh distribution is often used to model nonnegative magnitudes, such as wind speeds, wave heights, or signal amplitudes, when the underlying quantity can be viewed as the magnitude of a two-dimensional random vector. The probability density function for this distribution is

$$f(x;\theta) = \frac{x}{\theta}\exp\left(-\frac{x^2}{2\theta}\right), \qquad x > 0,\ \theta > 0.$$

Let $X_1, \ldots, X_n$ be a random sample from this distribution. For parts (d) and (e) it may help to know that

$$\mathbb{E}_\theta(X^2) = 2\theta, \qquad \mathbb{E}_\theta(X^4) = 8\theta^2.$$

(a) Write down the log-likelihood function for this model. [1 mark]
Solution. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{x_i}{\theta}\exp\left(-\frac{x_i^2}{2\theta}\right), \qquad \theta > 0.$$

Therefore

$$L(\theta) = \left(\prod_{i=1}^{n} x_i\right)\theta^{-n}\exp\left(-\frac{1}{2\theta}\sum_{i=1}^{n} x_i^2\right).$$

Hence the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n}\log x_i - n\log\theta - \frac{1}{2\theta}\sum_{i=1}^{n} x_i^2.$$

(b) Write down the score function, that is the derivative of the log-likelihood function with respect to θ. [1 mark]
Solution. Differentiating the log-likelihood gives

$$U(\theta) = \frac{d\ell(\theta)}{d\theta} = -\frac{n}{\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n} x_i^2.$$

(c) Determine the maximum likelihood estimator of θ. Make sure you establish that you have the global maximum of the likelihood. [2 marks]
Solution. Setting the score equal to zero gives

$$-\frac{n}{\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n} x_i^2 = 0.$$

The only solution to this equation is

$$\hat{\theta} = \frac{1}{2n} \sum_{i=1}^{n} X_i^2.$$

To check that this is the global maximum, note that

$$U(\theta) = \frac{S - 2n\theta}{2\theta^2}, \qquad S = \sum_{i=1}^{n} x_i^2,$$

which is positive for $\theta < S/(2n)$ and negative for $\theta > S/(2n)$. Hence $\hat{\theta} = S/(2n)$ is the global maximiser of the likelihood.

(d) Determine the bias of the maximum likelihood estimator of θ. [2 marks]

Solution. Since

$$\hat{\theta} = \frac{1}{2n} \sum_{i=1}^{n} X_i^2,$$

we have

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{1}{2n} \sum_{i=1}^{n} \mathbb{E}_\theta(X_i^2).$$

Using $\mathbb{E}_\theta(X^2) = 2\theta$,

$$\mathbb{E}_\theta(\hat{\theta}) = \frac{1}{2n}\, n(2\theta) = \theta.$$

Therefore

$$\text{Bias}(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta = 0.$$

Hence the maximum likelihood estimator is unbiased.

(e) Determine the mean squared error of the maximum likelihood estimator of θ. [3 marks]

Solution. Since $\hat{\theta}$ is unbiased,

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}).$$

Now

$$\text{Var}(X^2) = \mathbb{E}_\theta(X^4) - \{\mathbb{E}_\theta(X^2)\}^2 = 8\theta^2 - (2\theta)^2 = 4\theta^2.$$

Therefore

$$\text{Var}(\hat{\theta}) = \text{Var}\left(\frac{1}{2n} \sum_{i=1}^{n} X_i^2\right) = \frac{1}{4n^2} \sum_{i=1}^{n} \text{Var}(X_i^2) = \frac{1}{4n^2} n(4\theta^2) = \frac{\theta^2}{n}.$$

Thus

$$\text{MSE}(\hat{\theta}) = \frac{\theta^2}{n}.$$

(f) Is the maximum likelihood estimator of θ a consistent estimator? Justify your answer. [1 mark]

Solution. As the MSE goes to zero as $n \to \infty$, the maximum likelihood estimator is consistent.

2. (10 marks) The zero-truncated Poisson distribution is used to model count data where only positive counts can be observed, or where zero counts are excluded from the sample. The probability mass function for this distribution is

$$f(x;\theta) = \frac{\theta^x}{x!\{\exp(\theta)-1\}}, \qquad x \in \{1,2,\ldots\},\ \theta > 0.$$

Let $X_1, \ldots, X_n$ be a random sample from the zero-truncated Poisson distribution.

(a) Write down the log-likelihood function for this model. [1 mark]

Solution. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{\theta^{x_i}}{x_i!\{\exp(\theta)-1\}}.$$

Hence

$$L(\theta) = \frac{\theta^{\sum_{i=1}^n x_i}}{\{\prod_{i=1}^n x_i!\}\,\{\exp(\theta)-1\}^n}.$$

Therefore the log-likelihood is

$$\ell(\theta) = \left(\sum_{i=1}^{n} x_i\right)\log\theta - \sum_{i=1}^{n}\log(x_i!) - n\log\{\exp(\theta)-1\}.$$

(b) Write down the score function and show that the maximum likelihood estimator must satisfy the equation $g(\theta) = \bar{X}$, where $g : (0,\infty) \to (0,\infty)$ is the function defined by

$$g(\theta) = \frac{\theta}{1-\exp(-\theta)}$$

and $\bar{X} = n^{-1}\sum_{j=1}^n X_j$. [2 marks]

Solution. Differentiating with respect to θ gives

$$U(\theta) = \frac{d\ell(\theta)}{d\theta} = \frac{\sum_{i=1}^n x_i}{\theta} - n\frac{\exp(\theta)}{\exp(\theta)-1}.$$

Equivalently,

$$U(\theta) = n\left\{\frac{\bar{X}}{\theta} - \frac{1}{1-\exp(-\theta)} = \frac{n}{\theta}\left\{\bar{X} - g(\theta)\right\}\right\}.$$

(c) Show that the function g is strictly increasing on $(0,\infty)$. [3 marks]

Solution. Write

$$g(\theta) = \frac{\theta}{1-\exp(-\theta)}.$$

Differentiating gives

$$g'(\theta) = \frac{1-\exp(-\theta)-\theta\exp(-\theta)}{\{1-\exp(-\theta)\}^2}.$$

The denominator is positive for $\theta > 0$. The numerator can be written as

$$1-(1+\theta)\exp(-\theta).$$

From the Taylor expansion of the exponential function, we see $\exp(\theta) > 1+\theta$ for $\theta > 0$. It follows that

$$(1+\theta)\exp(-\theta) < 1.$$

Therefore the numerator is positive. Hence

$$g'(\theta) > 0, \qquad \theta > 0,$$

and so g is strictly increasing on $(0, \infty)$.

(d) When does the maximum likelihood estimator exist? [3 marks]

Solution. We need to solve

$$\bar{X} = g(\theta), \qquad \theta > 0.$$

Take the limit as $\theta \to 0$:

$$\begin{aligned} \lim_{\theta \to 0^+} g(\theta) &= \lim_{\theta \to 0^+} \frac{1}{\exp(-\theta)} && \text{[L'hopitals rule]} \\ &= 1 \end{aligned}$$

And the limit as $\theta \to \infty$:

$$\lim_{\theta \to \infty} g(\theta) = \lim_{\theta \to \infty} \frac{\theta}{1 - \exp(-\theta)} = \infty,$$

since $\lim_{\theta \to \infty}(1 - \exp(-\theta)) = 1$.
As g is strictly increasing and continuous, the range of g on $(0, \infty)$ is $(1, \infty)$. Therefore a finite MLE exists if and only if

$$\bar{X} > 1.$$

Since each observation is a positive integer, $\bar{X} = 1$ occurs exactly when all observations are equal to one. In that case there is no finite solution in $(0, \infty)$.

(e) Assuming the maximum likelihood estimator exists, is it unique? Justify your answer. [1 mark]

Solution. Yes. If the MLE exists, then $\bar{X} > 1$. Suppose there were two solutions θ_1 and θ_2 to $g(\theta) = \bar{X}$. Without loss of generality, assume $\theta_1 < \theta_2$. Since g is strictly increasing, $g(\theta_1) < g(\theta_2)$. This is a contradiction. Hence, the MLE is unique when it exists.

3. (10 marks)The Lomax distribution is a heavy-tailed distribution often used to model positive-valued data such as insurance losses, lifetimes, waiting times, or other skewed data with occasional large observations. The probability density function for this distribution is

$$f(x;\theta) = \frac{\alpha}{\lambda}\left(1+\frac{x}{\lambda}\right)^{-(\alpha+1)}, \qquad x>0, \quad \theta=(\alpha,\lambda)\in(0,\infty)^2.$$

The mean and variance of the Lomax distribution are

$$\mathbb{E}_\theta(X) = \begin{cases} \dfrac{\lambda}{\alpha-1}, & \alpha>1, \\ \infty, & 0<\alpha\le 1, \end{cases}$$

and

$$\mathrm{Var}_\theta(X) = \begin{cases} \dfrac{\lambda^2\alpha}{(\alpha-1)^2(\alpha-2)}, & \alpha>2, \\ \infty, & 0<\alpha\le 2. \end{cases}$$

Let $X_1,\ldots,X_n$ be a random sample from this distribution.

(a) Identify one difficulty with applying the method of moments for this distribution. [1 mark]

Solution. One difficulty is that the required theoretical moments do not always exist. The mean is finite only when $\alpha > 1$, and the variance is finite only when $\alpha > 2$. Therefore a method of moments estimator based on matching the sample mean and sample variance is only meaningful under the assumption that $\alpha > 2$. In addition, for heavy-tailed data, the sample mean and especially the sample variance can be unstable in finite samples.

(b) Assuming the relevant moments exist and are finite, provide expressions for the method of moments estimator of

$$\theta = (\alpha,\lambda).$$

Write your answer in terms of the sample mean and sample variance. You need to identify any condition(s) required for the method of moments estimator to give a value in the parameter space $(0,\infty)^2$. [4 marks]

Solution. Let

$$m_1 = \frac{1}{n}\sum_{i=1}^n X_i \quad \text{and} \quad m_2 = \frac{1}{n}\sum_{i=1}^n X_i^2.$$

Note that $\mathbb{E}_\theta(X^2) = \mathrm{Var}_\theta(X) + \mathbb{E}_\theta(X)^2$. The method of moments equations are

$$m_1 = \frac{\lambda}{\alpha-1},$$

and

$$m_2 = \frac{\lambda^2\alpha}{(\alpha-1)^2(\alpha-2)} + \left(\frac{\lambda}{\alpha-1}\right)^2 \qquad \Longrightarrow \qquad m_2 - m_1^2 = \frac{\lambda^2\alpha}{(\alpha-1)^2(\alpha-2)}.$$

From the first equation,

$$\lambda = m_1(\alpha-1).$$

Substituting this into the second equation gives

$$m_2 - m_1^2 = \frac{m_1^2(\alpha-1)^2\alpha}{(\alpha-1)^2(\alpha-2)} = \frac{m_1^2\alpha}{\alpha-2}.$$

Therefore

$$(m_2 - m_1^2)(\alpha - 2) = m_1^2\alpha.$$

Rearranging,

$$\alpha(m_2 - 2m_1^2) = 2(m_2 - m_1^2).$$

Solving the above equation and noting that $(\frac{n-1}{n})S^2 = m_2 - m_1^2$,

$$\hat{\alpha}_{\text{MOM}} = \frac{2(m_2 - m_1^2)}{m_2 - 2m_1^2} = \frac{2(\frac{n-1}{n})S^2}{(\frac{n-1}{n})S^2 - \bar{X}^2},$$

and

$$\hat{\lambda}_{\text{MOM}} = \bar{X}(\hat{\alpha}_{\text{MOM}} - 1) = \frac{\bar{X}((\frac{n-1}{n})s^2 + \bar{X}^2)}{(\frac{n-1}{n})s^2 - \bar{X}^2}.$$

provided $(\frac{n-1}{n})S^2 > \bar{X}^2$.

(c) Write down the log-likelihood function for this model. [1 mark]

Solution. The likelihood is

$$L(\alpha, \lambda) = \prod_{i=1}^{n} \frac{\alpha}{\lambda}\left(1 + \frac{x_i}{\lambda}\right)^{-(\alpha+1)}.$$

Therefore the log-likelihood is

$$\ell(\alpha, \lambda) = n\log\alpha - n\log\lambda - (\alpha + 1)\sum_{i=1}^{n}\log\left(1 + \frac{x_i}{\lambda}\right),$$

for $\alpha > 0$ and $\lambda > 0$.

(d) The file `lomax.csv` provides 1000 observations. In R or Python, write a suitable function to evaluate the log-likelihood for this model. Use `optim` in R (or an equivalent numerical optimiser in Python) to find the maximum likelihood estimate for this data. You may want to use the method of moments estimator from part (b) as an initial value.
You must show your code and the output from `optim` (or equivalent optimiser in Python).

[4 marks]

Note: You can read the data into R using the following code:

```
data <- scan("lomax.csv")
```

assuming that the file `lomax.csv` is in the working directory.

Solution.

```
data <- scan("lomax.csv")

# Method of moments starting values
xbar <- mean(data)
s2 <- var(data)
n <- length(data)

alpha0 <- 2 * (n/(n-1))* s2 / ((n/(n-1))* s2 - xbar^2)
lambda0 <- xbar * (alpha0 - 1)
```

```
MoM <- c(alpha0, lambda0)
names(MoM) <- c("alpha", "lambda")
MoM
    alpha    lambda
 4.520276 35.965624


# Negative log likelihod function for the Lomax distribution
negloglik <- function(theta, data) {
  alpha <- theta[1]
  lambda <- theta[2]
  loglik <- length(data) * log(alpha) - length(data) * log(lambda) -
    (alpha + 1) * sum(log(1 + data / lambda))
  return(-loglik)
}

# Find the MLE by optimising
fit <- optim(par = MoM,
             fn = negloglik,
             method = "L-BFGS-B",
             lower = c(0,0),
             data=data)

mle <- fit$par
names(mle) <- c("alpha", "lambda")
mle

    alpha    lambda
 4.857471 39.301545
```

4. (10 marks) Let $X_1, \ldots, X_n$ be a random sample from a $\mathsf{Binomial}(2, p)$ distribution, where $p \in [0, 1]$. We are interested in estimating p^2, the probability of two successes from two trials.

   (a) Define random variables $Y_1, \ldots, Y_n$ by

$$Y_j = \begin{cases} 1, & X_j = 2, \\ 0, & X_j \neq 2. \end{cases}$$

   Show that $n^{-1} \sum_{j=1}^{n} Y_j$ is an unbiased estimator of p^2. [2 marks]
   Solution. Since $X_j \sim \mathsf{Binomial}(2, p)$,

$$\mathbb{P}(X_j = 2) = \binom{2}{2} p^2 (1-p)^0 = p^2.$$

   Therefore

$$Y_j \sim \mathsf{Bernoulli}(p^2),$$

   so

$$\mathbb{E}(Y_j) = p^2.$$

   Hence

$$\mathbb{E}\left(\frac{1}{n}\sum_{j=1}^{n} Y_j\right) = \frac{1}{n}\sum_{j=1}^{n} \mathbb{E}(Y_j) = p^2.$$

   Thus $n^{-1} \sum_{j=1}^{n} Y_j$ is unbiased for p^2.

   (b) Determine the mean squared error of the estimator in part (a). [2 marks]
   Solution. Since the estimator in part (a) is unbiased, its MSE is its variance. Since $Y_j \sim \mathsf{Bernoulli}(p^2)$,

$$\mathrm{Var}(Y_j) = p^2(1 - p^2).$$

   Therefore

$$\mathrm{MSE}\left(\frac{1}{n}\sum_{j=1}^{n} Y_j\right) = \mathrm{Var}\left(\frac{1}{n}\sum_{j=1}^{n} Y_j\right) = \frac{p^2(1-p^2)}{n}.$$

   (c) It is known that the maximum likelihood estimator of p is $\hat{p} = \frac{1}{2n}\sum_{j=1}^{n} X_j$. Determine the maximum likelihood estimator of p^2. [1 mark]
   Solution. By the invariance property of maximum likelihood estimators, the MLE of p^2 is obtained by substituting $\hat{p}$ into p^2. Thus

$$\widehat{p^2}_{\mathrm{MLE}} = \hat{p}^{\,2} = \left(\frac{1}{2n}\sum_{j=1}^{n} X_j\right)^2.$$

   (d) Determine the bias of the maximum likelihood estimator of p^2 from part (c). [3 marks]

   Solution. Let

$$S = \sum_{j=1}^{n} X_j.$$

   Since each $X_j \sim \mathsf{Binomial}(2, p)$, we have

$$S \sim \mathsf{Binomial}(2n, p).$$

Therefore

$$\hat{p} = \frac{S}{2n}, \qquad \mathbb{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{p(1-p)}{2n}.$$

Now

$$\mathbb{E}(\hat{p}^2) = \mathrm{Var}(\hat{p}) + \{\mathbb{E}(\hat{p})\}^2.$$

Hence

$$\mathbb{E}(\hat{p}^2) = \frac{p(1-p)}{2n} + p^2.$$

Therefore the bias is

$$\mathrm{Bias}(\hat{p}^2) = \mathbb{E}(\hat{p}^2) - p^2 = \frac{p(1-p)}{2n}.$$

Thus the MLE of p^2 is biased upward.

(e) For large n, the mean squared error of the maximum likelihood estimator of p^2 is approximately

$$\frac{2p^3(1-p)}{n}.$$

Using this approximation, compare the mean squared error of the maximum likelihood estimator with the mean squared error of the estimator in part (a). Which estimator is preferred under this criterion? [2 marks]

Solution. From part (b), the MSE of the unbiased estimator is

$$\mathrm{MSE}\left(\frac{1}{n}\sum_{j=1}^{n} Y_j\right) = \frac{p^2(1-p^2)}{n}.$$

The approximate MSE of the MLE is

$$\mathrm{MSE}(\hat{p}^2) \approx \frac{2p^3(1-p)}{n}.$$

Compare the ratio:

$$\frac{\mathrm{MSE}(\hat{p}^2)}{\mathrm{MSE}\left(n^{-1}\sum_{j=1}^{n} Y_j\right)} \approx \frac{2p^3(1-p)}{p^2(1-p^2)}.$$

Since $1 - p^2 = (1-p)(1+p)$, this becomes

$$\frac{2p^3(1-p)}{p^2(1-p)(1+p)} = \frac{2p}{1+p}.$$

For $0 < p < 1$,

$$\frac{2p}{1+p} < 1.$$

Therefore, using the large-sample MSE approximation, the MLE has smaller MSE than the unbiased estimator from part (a). Under this criterion, the MLE is preferred.

Only for STAT2904

5. (10 marks) Let m be known. Suppose A is an unknown non-empty subset of $\{1, \ldots, m\}$. Let $X_1, \ldots, X_n$ be independent random variables with

$$\mathbb{P}_A(X = x) = \begin{cases} 1/|A|, & x \in A, \\ 0, & x \notin A. \end{cases}$$

(a) Write down the likelihood function for A. [2 marks]
Solution. Let

$$B = \{x_1, \ldots, x_n\}$$

denote the set of distinct observed values. The joint probability of the observed sample is

$$L(A) = \prod_{i=1}^{n} \mathbb{P}_A(X_i = x_i).$$

If every observed value belongs to A, then each probability is $1/|A|$. If at least one observed value is not in A, the likelihood is zero. Therefore

$$L(A) = |A|^{-n} \mathbb{I}(B \subseteq A),$$

where A is a non-empty subset of $\{1, \ldots, m\}$.

(b) Find the maximum likelihood estimator of A. [2 marks]
Solution. The likelihood is positive only if $B \subseteq A$. Among all subsets A that contain B, the likelihood is

$$L(A) = |A|^{-n}.$$

This is maximised by making $|A|$ as small as possible. The smallest subset containing all observed values is B itself. Therefore

$$\hat{A} = B = \{X_1, \ldots, X_n\}.$$

That is, the MLE of A is the set of distinct observed values.

(c) Let $K = |A|$, the cardinality of set A. Use the invariance property to find the maximum likelihood estimator of K. [1 mark]
Solution. By the invariance property, the MLE of $K = |A|$ is

$$\hat{K} = |\hat{A}|.$$

Since $\hat{A}$ is the set of distinct observed values,

$$\hat{K} = |\{X_1, \ldots, X_n\}|.$$

Thus $\hat{K}$ is the number of distinct values observed in the sample.

(d) Is the maximum likelihood estimator of K biased? Justify your answer. [3 marks]
Solution. Yes, except in the trivial case $K = 1$.
For each element $a \in A$, define

$$I_a = \mathbb{I}\{a \text{ is observed at least once in the sample}\}.$$

Then

$$\hat{K} = \sum_{a \in A} I_a.$$

For a fixed $a \in A$,

$$\mathbb{P}(a \text{ is not observed}) = \left(1 - \frac{1}{K}\right)^n.$$

Hence

$$\mathbb{P}(a \text{ is observed at least once}) = 1 - \left(1 - \frac{1}{K}\right)^n.$$

Therefore

$$\mathbb{E}(\hat{K}) = \sum_{a \in A} \mathbb{E}(I_a) = K\left\{1 - \left(1 - \frac{1}{K}\right)^n\right\}.$$

Thus

$$\text{Bias}(\hat{K}) = \mathbb{E}(\hat{K}) - K = -K\left(1 - \frac{1}{K}\right)^n.$$

This is negative when $K > 1$, so $\hat{K}$ is biased downward.

(e) Is the maximum likelihood estimator of A consistent as $n \to \infty$, with m and A fixed? Justify your answer. [2 marks]

Solution. Yes. The estimator $\hat{A}$ is equal to A exactly when every element of A has been observed at least once. The probability that a particular element $a \in A$ is not observed in n draws is

$$\left(1 - \frac{1}{K}\right)^n, \qquad K = |A|.$$

By the union bound,

$$\begin{aligned}\mathbb{P}(\hat{A} \neq A) &= \mathbb{P}\left(\bigcup_{a \in A} \text{element } a \text{ is not sampled in } n \text{ draws}\right) \\ &\leq \sum_{a \in A}\left(1 - \frac{1}{K}\right)^n = K\left(1 - \frac{1}{K}\right)^n.\end{aligned}$$

Since K is fixed and

$$\left(1 - \frac{1}{K}\right)^n \to 0 \qquad \text{as } n \to \infty,$$

it follows that

$$\mathbb{P}(\hat{A} \neq A) \to 0.$$

Hence

$$\mathbb{P}(\hat{A} = A) \to 1,$$

so $\hat{A}$ is consistent for A.